

ABC Call Volume Trend Analysis

Project Description:

ABC Call Volume Trend Analysis refers to the act of reviewing and analyzing the fluctuations, patterns, and trends in call volume specific to the "ABC" organization or system.

ABC company analysis might include:

- Identifying specific trends in call volume, such as peak hours (e.g., morning, afternoon), high-demand days (e.g., Mondays or weekends)
- Helping ABC optimize staffing and operational strategies based on expected call traffic, ensuring that the right number of agents are available during peak times without overstaffing during slower periods.

In this project, we'll be diving into the world of Customer Experience (CX) analytics, specifically focusing on the inbound calling team of a company.

We'll be provided with a dataset that spans 23 days and includes various details such as the agent's name and ID, the queue time (how long a customer had to wait before connecting with an agent), the time of the call, the duration of the call, and the call status (whether it was abandoned, answered, or transferred).

The Customer Experience (CX) team is essential to a company, responsible for analyzing customer feedback and data to extract valuable insights. They oversee customer experience programs, manage internal communications, map customer journeys, and handle customer data, among other tasks.

In this project, you'll apply analytical skills to analyze the CX team's call volume trends and extract valuable insights.

Assumptions:

Working days: 6 days a week

Each agent : 4 unplanned leaves per month;

An agent's total working hours :9 hours

Break (lunch & snacks): 1.5 hours

Actual working hours: 7.5 hours

Call working hours is 60% of actual working hours :4.5 hours

Total number of days in a month is 30.

Approach

As I got the raw data set, my first automatic instinct is to clean the data, so that we can move forward with the analytical processes

The approach includes:

Data understanding:

Total rows: 117989

Total columns: 13

Null Values: 47877

Unusual values: In Agent_Name & Agent_ID 34198 cells

Data Cleaning:

To count blank values and delete it, used “count blank” function to know blank cells column wise. “Wrapped_By” column found to be had 47877 blank cells to handle that Replaced missing value with Abandoned Call where call status labelled as Abandon and for remaining used Mode function to replace missing data, Replaced with Agent because it has highest mode

Then we found some unusual values in Agent_Name & Agent_ID as “#N/A” I counted them overall with countif function and found 34198 cells with that, replaced them all with Abandoned call cause after looking at it thoroughly found that these all were referring with call status were abandon too

Then for data analytics used basic statistics for call volume and other relevant metrics like Average Call Duration, Call Volume Analysis, Manpower Planning, Night Shift Manpower Planning

Visualization Tools: PivotTables, Line/column Charts, Conditional Formatting

By following this approach, I was able to efficiently analyze call volume trends for ABC Company, identify patterns, forecast future volumes, and provide data-driven recommendations to optimize call center operations.

Tech – stack used:

Microsoft Excel: To organize, analyze, and present data in a simple yet powerful manner

Microsoft Word: For documenting, editing, and sharing the process what I've done throughout the project

Canva & Microsoft power point: To use the proper templates and presenting project very appealing and interesting

Insights:

1. Average Call Duration: Determine the average duration of all incoming calls received by agents. This should be calculated for each time bucket.

My Task: What is the average duration of calls for each time bucket?

Ans: After I've cleaned the dataset to easily analyse the data and provide informative insights

First, I've inserted cleaned data into pivot table and then inserted call_status in to filters, & filtered into only answered

Time_bucket in to rows &

call_seconds(s) into values, took average of call seconds

with the data, inserted it with clustered column chart to visually represented "Average Duration of Calls for Each Time Bucket"

Maximum duration was found during 10-11,18-19,19-20,20-21(i.e. 203)

Minimum duration was found during 12-13,13-14,14-15(i.e. 193,194,195)

Total Average call duration in morning shift is 199 seconds

2.Call Volume Analysis: Visualize the total number of calls received.

This should be represented as a graph or chart showing the number of calls against time. Time should be represented in buckets (e.g., 1-2, 2-3, etc.).

My Task: Can you create a chart or graph that shows the number of calls received in each time bucket?

Ans: Inserted cleaned data into pivot table and then inserted call_status in to filters, then selected all

Time_bucket in to rows &

call_seconds(s) into values, took countof call seconds (titled as calls Received)

with the data, inserted it with Bar chart to visually represented & also calculated percentage so that u could understand how much percentage goes

within 1hr duration inserted it with clustered column chart to know how many calls received during different Time_Bucket

Maximum calls received during (10-11,11-12,12-13,13-14)

Minimum calls received during (19-20,20-21)

Total average call received is 117988

3.Manpower Planning: The current rate of abandoned calls is approximately 30%. Propose a plan for manpower allocation during each time bucket (from 9 am to 9 pm) to reduce the abandon rate to 10%. In other words, you need to calculate the minimum number of agents required in each time bucket to ensure that at least 90 out of 100 calls are answered.

My Task: What is the minimum number of agents required in each time bucket to reduce the abandon rate to 10%?

Ans: To reduce the abandoned call rate to 10%, we need to ensure that at least 90% of calls are answered in a timely manner. This means the number of agents available must be sufficient to handle the incoming calls, based on their average handle time and the incoming call volume.

With the cleaned data, I've inserted pivot table with call_status into rows & customer_phone_no in to values to know number of calls answered, abandoned & transferred

And then with the data I've inserted other pivot table with the call_status into columns, Time_Bucket into rows and took the count of customer_phone_no

And with assumptions presented, Took the data provided took only abandon and answered calls calculated per 23 days as well as per day too

With ROUND (number, num_digits) excel function for abandon and answered calls calculated total with sum excel function and calculated percentage

Inserted all the data into clustered column to understand graphically

Calculated Required Daily Call Volume to Achieve 90% Performance with total *0.9 as we need 90% for all time durations

Calculated agents available with answered call / Performance Agent's call handling capacity per hour for all time durations

And also Calculated Required Number of Agents to answer 90% call with Required Daily Call Volume to Achieve 90% /Performance Agent's call handling capacity per hour for all time durations

Inserted data to understand it more precisely visually to line chart to know how many agents are present and how many more required

Maximum Agents required during (9-10,10-11,11-12)

No extra Man power required during (15-16,16-17,17-18) as these time slots have required agents as present

More 55 agents required to answer 90% of daily incoming calls

4.Night Shift Manpower Planning: Customers also call ABC Insurance Company at night but don't get an answer because there are no agents available. This creates a poor customer experience. Assume that for every 100 calls that customers make between 9 am and 9 pm, they also make 30 calls at night between 9 pm and 9 am. The distribution of these 30 calls is as follows:

My Task: Propose a manpower plan for each time bucket throughout the day, keeping the maximum abandon rate at 10%

Ans: calculating Total No of Calls Per Night with Total No of Calls Per Day*0.3 which is 30% of Total calls

And as been provided us with the distribution of 30 calls during time slots took the percentage and calculated no of calls with distribution of calls/ Total No of Calls Per Night of all time slots calculated number of agents required for night with no of calls / Agent's call handling capacity per hour and inserted all data into clustered column to understand more clearly

Agents require more during morning time that would be 6-7am,7-8am,8-9am

Minimum agents required during 1-2am,2-3am,3-4am,4-5am

Total 76 agents required to handle calls at night

And finally made a schedule for how many agents required to answer 90% calls during morning & night time slots & visually represented with Line chart

From the chart its very clear that we need a greater number of agents during morning shift to answer 90% calls

Excel sheet of all tasks: [ABC Call Volume Trend Analysis.xlsx](#)